INCH-POUND

MIL-C-83723/35E
10 November 1988
SUPERSEDING
MIL-C-83723/35D(USAF)
3 December 1979

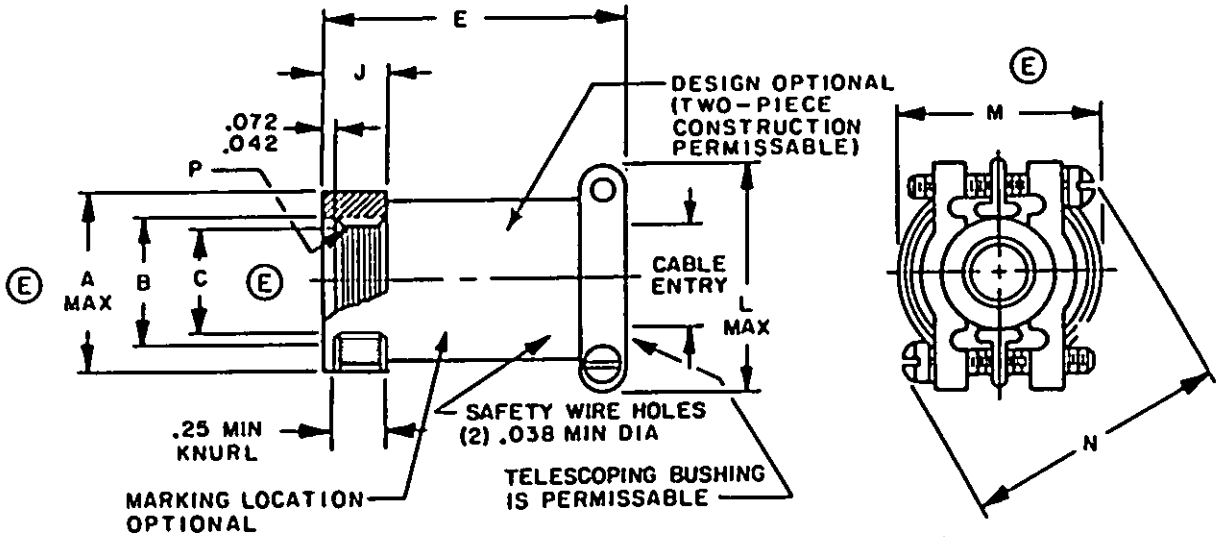
MILITARY SPECIFICATION SHEET

CONNECTOR, ELECTRIC, BACKSHELL, STRAIGHT W/O STRAIN RELIEF, AND WITH MECHANICAL STRAIN RELIEF, STRAIGHT OR RIGHT ANGLE THREADED COUPLING SERIES II

INACTIVE FOR NEW DESIGN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the backshell described herein shall consist of this specification and the latest issue of MIL-C-83723.



Inches	mm
.038	0.97
.042	1.07
.072	1.83
.25	6.35

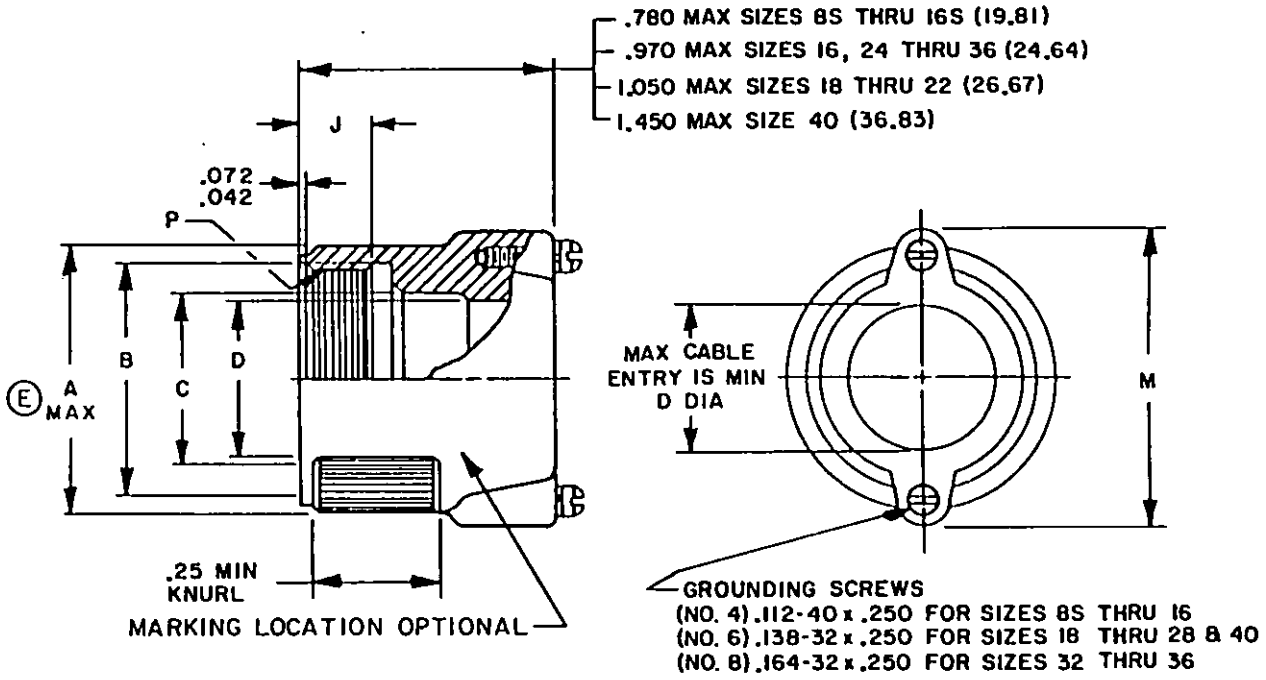
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Millimeters are in parentheses.
4. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals, ±.010 (0.25 mm) for two place decimals, and angular tolerances X^{o} = ±1° 0', $X^{o}X'$ = ±30'.
5. Dimensions apply after plating.
6. Remove all burrs and sharp edges.

FIGURE 1. Type S - Straight clamp.

Ⓔ denotes changes

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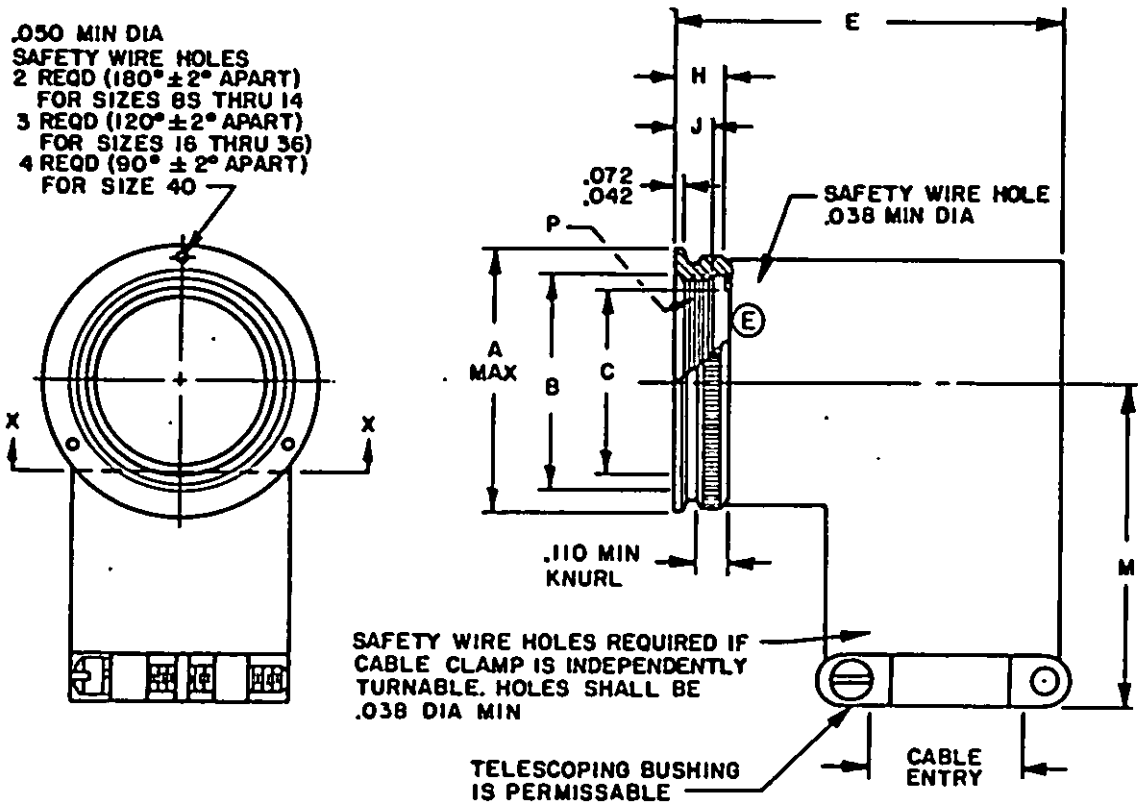


Inches	mm
.042	1.07
.072	1.83
.112	2.84
.138	3.51
.164	4.17
.250	6.35

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Millimeters are in parentheses.
4. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals, ±.010 (0.25 mm) for two place decimals, and angular tolerances X^{o} = ±1° 0', $X^{o}X'$ = ±30'.
5. Dimensions apply after plating.
6. Remove all burrs and sharp edges.

FIGURE 2. Type N - no clamp.

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Inches	mm
.038	0.97
.042	1.07
.050	1.27
.072	1.83
.110	2.79

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Millimeters are in parentheses.
4. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals, ±.010 (0.25 mm) for two place decimals, and angular tolerances X° = ±1° 0', X°X' = ±30'.
5. Dimensions apply after plating.
6. Remove all burrs and sharp edges.

FIGURE 3. Type A - 90° clamp.

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TABLE I. Dimensions figures 1 and 2.

Shell size part number designation	A Max Dia	B Dia	C Dia	D Dia	E Max	J Min	L Max	M Max	N Max	P Thread Class 2B	Cable entry 3/ Fig.1 Max	Cable entry 3/ Fig.1 Min
08	.585	.450 .440	.291 .281	.257 .245	1.290 (32.77)	.285	.843	.900 (22.86)	1.075	.438-28 UNEF	.235	.125
10	.645	.513 .503	.351 .341	.310 .300	"	"	.843	.900 (22.86)	1.075	.500-28 "	.235	.125
11	.730	.575 .565	.420 .410	.390 .380	"	"	.906	.980 (24.89)	1.140	.562-24 "	.297	.170
13 1/	.770	.640 .628	.471 .461	.415 .403	"	"	1.085	.995 (25.27)	1.140	.625-24 "	.297	.170
15 2/	.910	.765 .753	.575 .564	.497 .485	1.368 (34.75)	"	1.093	1.160 (29.46)	1.360	.750-20 "	.422	.235
17	1.035	.890 .878	.700 .690	.616 .604	1.430 (36.32)	"	1.187	1.260 (32.00)	1.520	.875-20 "	.547	.295
16	1.035	.890 .878	.700 .690	.616 .604	1.587 (40.31)	.440	1.275	1.260 (32.00)	1.520	.875-20 "	.547	.295
18	1.170	1.015 1.003	.817 .807	.735 .725	1.695 (43.05)	"	1.475	1.455 (36.96)	1.600	1.000-20 "	.610	.325
20	1.300	1.140 1.128	.953 .943	.875 .865	1.695 (43.05)	"	1.500	1.580 (40.13)	1.890	1.125-18 "	.735	.380
22	1.420	1.265 1.255	1.083 1.073	.983 .973	1.695 (43.05)	"	1.710	1.580 (40.13)	1.890	1.250-18 "	.735	.380
24	1.535	1.390 1.378	1.190 1.180	1.095 1.085	1.931 (49.05)	"	1.719	1.890 (48.01)	2.145	1.375-18 "	.922	.500
28	1.800	1.640 1.628	1.410 1.400	1.255 1.245	1.931 (49.05)	"	2.040	1.890 (48.01)	2.145	1.625-18 "	.922	.500
32	2.035	1.890 1.878	1.664 1.654	1.525 1.515	1.869 (47.47)	"	2.332	2.215 (56.26)	2.550	1.875-16 UN	1.235	.705
36	2.270	2.140 2.128	1.875 1.862	1.760 1.750	1.869 (47.47)	"	2.312	2.405 (61.09)	2.889	2.125-16 UN	1.360	.725
40	2.780	2.395 2.385	2.208 2.198	2.020 1.865	2.875 (73.03)	.459	2.950	2.781 (70.64)	3.500	2.375-16 UN	1.630	.911

1/ Shell size 13 fits connector sizes 12 and 13.

2/ Shell size 15 fits connector sizes 14 and 15.

3/ "Cable entry" dimension is defined as the envelope area for the wire bundle. This dimension is not intended to define the clamp hardware limits.

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TABLE II. Dimensions figure 3.

Shell size part number designation	A Max Dia	B Dia	C Dia	E Max	H Max	J	M Max	P Thread Class 2A	Max weight lbs.	Cable entry 1/ Max	Cable entry 1/ Min
08	.805	.455 .440	.300 .279	1.187 (30.15)	.350	.285 .272	1.220	.438-28 UNEF	.025	.235	.102
10	.890	.520 .500	.360 .339	1.187 (30.15)	"	"	1.250	.500-28 "	.025	.235	.102
11	.975	.580 .560	.430 .408	1.312 (33.32)	"	"	1.280	.562-24 "	.033	.297	.140
13	1.005	.645 .625	.480 .459	1.312 (33.32)	"	"	"	.625-24 "	.030	"	"
12	1.005	.645 .625	.480 .459	1.468 (37.29)	.505	.441 .428	"	.625-24 "	.035	"	"
15	1.130	.770 .750	.584 .562	1.437 (36.50)	.355	.285 .272	1.405	.750-20 "	.040	.422	.195
14	1.130	.770 .750	.584 .562	1.594 (40.49)	.510	.441 .428	1.405	.750-20 "	.045	.422	.195
17	1.255	.895 .875	.710 .688	1.531 (38.89)	.350	.285 .272	1.529	.875-20 "	.045	.547	.255
16	1.255	.895 .875	.710 .688	1.687 (42.85)	.510	.441 .428	1.529	.875-20 "	.050	.547	.255
18	1.350	1.020 1.000	.827 .805	1.765 (44.83)	.490	"	1.593	1.000-20 "	.058	.610	.285
20	1.475	1.145 1.125	.962 .942	2.000 (50.80)	.495	"	1.655	1.125-18 "	.080	.735	.350
22	1.600	1.270 1.250	1.095 1.073	2.000 (50.80)	.495	"	1.718	1.250-18 "	.082	.735	.350
24	1.725	1.395 1.375	1.202 1.180	2.265 (57.53)	.488	.423 .410	1.894	1.375-18 "	.103	.922	.468
28	1.975	1.645 1.625	1.423 1.401	2.265 (57.53)	"	"	1.973	1.625-18 "	.110	.922	.468
32	2.225	1.895 1.875	1.674 1.652	2.578 (65.48)	"	"	2.187	1.875-16 UN	.145	1.235	.664
36	2.475	2.145 2.125	1.887 1.865	2.815 (71.50)	"	"	2.405	2.125-16 UN	.185	1.360	.694
40	2.750	2.390 2.375	2.125 2.117	5.800 (147.32)	.500	"	5.100	2.375-16 UN	---	1.630	.911

1/ "Cable entry" dimension is defined as the envelope area for the wire bundle. This dimension is not intended to define the clamp hardware limits.

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REQUIREMENTS:

(E) Material: Aluminum alloy in accordance with QQ-A-225, ASTM-B-85 Aluminum Alloy Die Castings, Stainless steel in accordance with QQ-P-35, Aluminum black anodize in accordance with MIL-A-8625.

Telescoping type bushing shall be capable of passing fluid and temperature requirements of the basic specification.

Example of part number:

M83723 / 35 A 22 R

Military number prefix

Slash sheet

Type

S: Straight clamp

A: 90° clamp

N: No clamp

Shell size

Class, finish

(E) Material and finish:

(E) R: Aluminum, 200°C, conductive finish

G: Stainless steel passivated, 200°C

(E) A: Aluminum black anodize, nonconductive, 200°C.

CONCLUDING MATERIAL

Custodians:
(E) Navy - AS
Air Force - 85

Review activities:
(E) Air Force - 11, 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
(E) DLA - ES

(Project 5935-3659)

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